

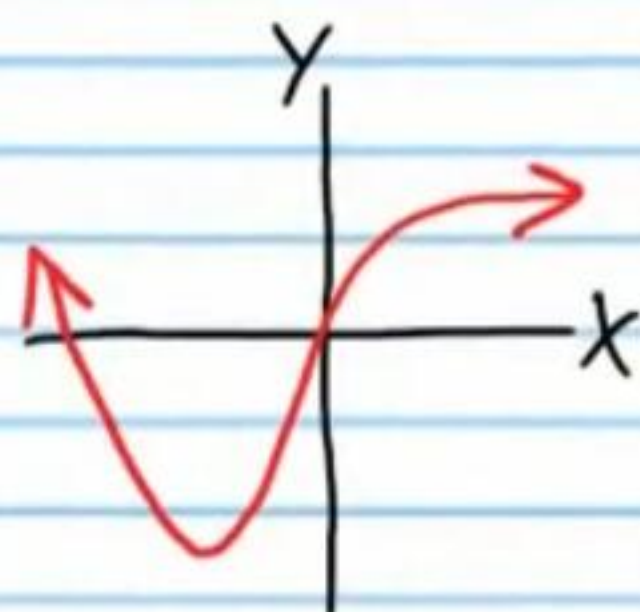
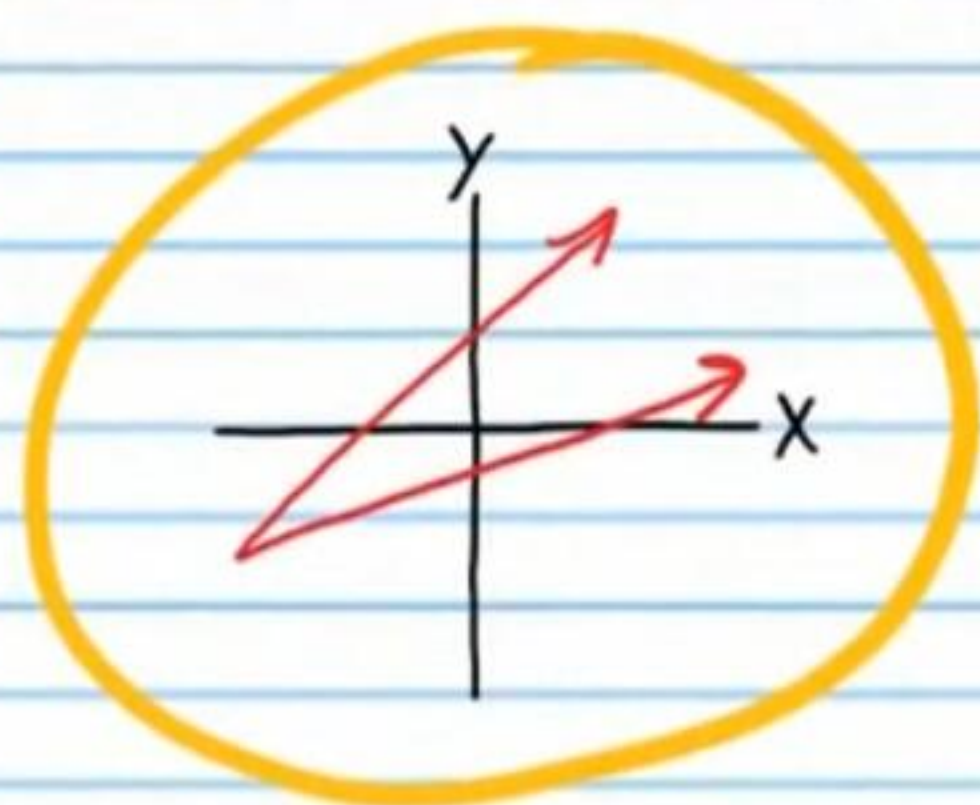
Practice Intermediate Algebra Final Test (Version 1)

- I) This is a cumulative exam.
- II) Write your name, the date, and the course title (i.e. Intermediate Algebra).
- III) Show all work.
- IV) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- V) You will need a calculator for this test.
- VI) You do not have to write the Roman numeral for each exponent rule that you use on this test.
- VII) Do not leave your answers with fraction exponents, negative exponents, zero power exponents or 1st power exponents.
- VIII) Each problem is worth eight points for a total of 200 points.

① Circle the graph or table that cannot represent a function.

X	3	4	3	8	7	10
Y	0	6	0	4	1	1

X	-1	-2	-3	-4	-5	-6
Y	-1	-2	-3	-3	-3	-3



② Multiply and write in $a+bi$ form.

$$(i+2)(3i-4) = 3i^2 - 4i + 6i - 8 = 3(-1) + 2i - 8 = -3 + 2i - 8 = -11 + 2i$$

③ Solve. Round to the nearest thousandth.

$$8e^{3x} + 2 = 5$$

$$8e^{3x} = 3$$

$$e^{3x} = \frac{3}{8}$$

$$3x = \log_e\left(\frac{3}{8}\right)$$

$$3x = \ln\left(\frac{3}{8}\right)$$

$$3x \approx -0.980829$$

$$x \approx -0.327$$

⑤ Solve.

$$2\log_x 5 - 3\log_x 2 = 1$$

$$\log_x 5^2 - \log_x 2^3 = 1$$

$$\log_x 25 - \log_x 8 = 1$$

$$\log_x \left(\frac{25}{8}\right) = 1$$

④ Simplify.

$$\left(\frac{x^3}{y^2}\right)^5 (x^6 \cdot x^3 \cdot b^0)$$

$$\frac{(x^3)^5}{(y^2)^5} (x^9 \cdot 1)$$

$$\frac{x^{15}}{y^{10}} (x^9)$$

$$\boxed{\frac{x^{24}}{y^{10}}}$$

$$x' = \frac{25}{8}$$

$$\boxed{x = \frac{25}{8}}$$

⑥ Solve.

$$\sqrt[3]{4} = \frac{8^x}{8^{2x}}$$

$$\sqrt[3]{2^2} = 8^{-x}$$

$$2^{\frac{2}{3}} = (2^3)^{-x}$$

$$2^{\frac{2}{3}} = 2^{-3x}$$

$$\frac{2}{3} = -3x$$

$$\boxed{-\frac{2}{9} = x}$$

⑦ Solve.

$$3^x \cdot 4^x = \frac{1}{144^{x-1}}$$

$$(3 \cdot 4)^x = \frac{1}{(12^2)^{x-1}}$$

$$12^x = \frac{1}{12^{2x-2}}$$

$$12^x = 12^{-(2x-2)}$$

$$12^x = 12^{-2x+2}$$

$$x = -2x + 2$$

$$3x = 2$$

$$\boxed{x = \frac{2}{3}}$$

⑧ Find an equation of the parabola with vertex (2,10) and point (-2,1).

$$y = a(x-h)^2 + k$$

$$1 = a(-2-2)^2 + 10$$

$$1 = a(-4)^2 + 10$$

$$1 = a(16) + 10$$

$$-9 = 16a$$

$$a = -\frac{9}{16}$$

$$\boxed{y = -\frac{9}{16}(x-2)^2 + 10}$$

⑨ If $f(x) = \ln x$ and $g(x) = |x+3|$, Find the following:

$$i) \left(\frac{f}{g}\right)(1) = \frac{f(1)}{g(1)} = \frac{\ln(1)}{|1+3|} = \frac{\log_e(1)}{|4|} = \frac{0}{4} = \boxed{0}$$

$$ii) (f \circ g)(x) = f(g(x)) = f(|x+3|) = \boxed{\ln |x+3|}$$

Find the values below if $h(x) = \begin{cases} 8x-2, & -6 < x < -2 \\ x^3, & -2 \leq x < 10 \end{cases}$

$$iii) h(1) = \boxed{\text{Undefined}}$$

$$iv) h(-2) = (-2)^3 = \boxed{-8}$$

$$v) h(3) = 3^3 = \boxed{27}$$

$$vi) h(-5) = 8(-5) - 2 = -40 - 2 = \boxed{-42}$$

vii) Write the domain of $h(x)$ in interval notation.

$$\boxed{(-6, 10)}$$

viii) Write the domain of $h(x)$ in set notation.

$$\boxed{\{x \mid -6 < x < 10\}}$$

⑩ Find the algebraic solution and the graphical solution.

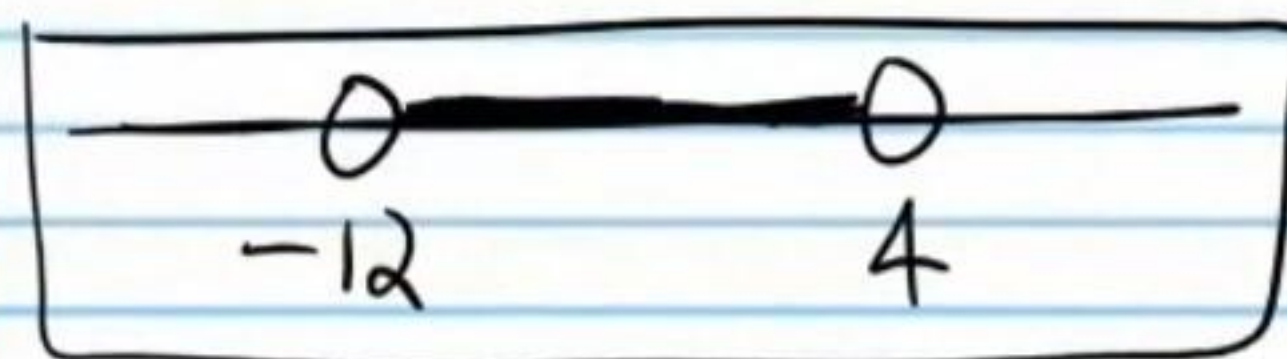
$$2 - 3\left|\frac{x}{4} + 1\right| > -4$$

$$-3\left|\frac{x}{4} + 1\right| > -6$$

$$\left|\frac{x}{4} + 1\right| < 2$$

$$\frac{x}{4} + 1 < 2 \text{ and } \frac{x}{4} + 1 > -2$$

$$\boxed{x < 4 \text{ and } x > -12}$$



⑪ Solve.

$$\log_3 10 = \log_3 (x-9) + \log_3 x$$

$$\log_3 10 = \log_3 [(x-9)x]$$

$$\log_3 10 = \log_3 (x^2 - 9x)$$

$$10 = x^2 - 9x$$

$$0 = x^2 - 9x - 10$$

$$0 = (x-10)(x+1)$$

$\swarrow \searrow$
 $\boxed{x=10}$ or $x \neq -1$

*Beware of
extraneous
solutions.

⑫ Solve.

$$-9 \log \left(\frac{2-x}{4x^2} \right) + 2 = 2$$

$$-9 \log_{10} \left(\frac{2-x}{4x^2} \right) + 2 = 2$$

$$-9 \log_{10} \left(\frac{2-x}{4x^2} \right) = 0$$

$$\log_{10} \left(\frac{2-x}{4x^2} \right) = 0$$

$$10^0 = \frac{2-x}{4x^2}$$

$$1 = \frac{2-x}{4x^2}$$

$$4x^2 \cdot 1 = \frac{2-x}{4x^2} \cdot 4x^2$$

$$4x^2 = 2-x$$

$$4x^2 + x - 2 = 0$$

$$4x^2 + x - 2 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-1 \pm \sqrt{1^2 - 4(4)(-2)}}{2(4)}$$

$$x = \frac{-1 \pm \sqrt{1+32}}{8}$$

$$x = \frac{-1 \pm \sqrt{33}}{8}$$

$\swarrow \searrow$
 $\boxed{x = \frac{-1 + \sqrt{33}}{8}}$ or $\boxed{x = \frac{-1 - \sqrt{33}}{8}}$

⑬ i) Graph the function below by performing multiple transformations on the graph of the parent function. *Graph each transformation and write its equation.*

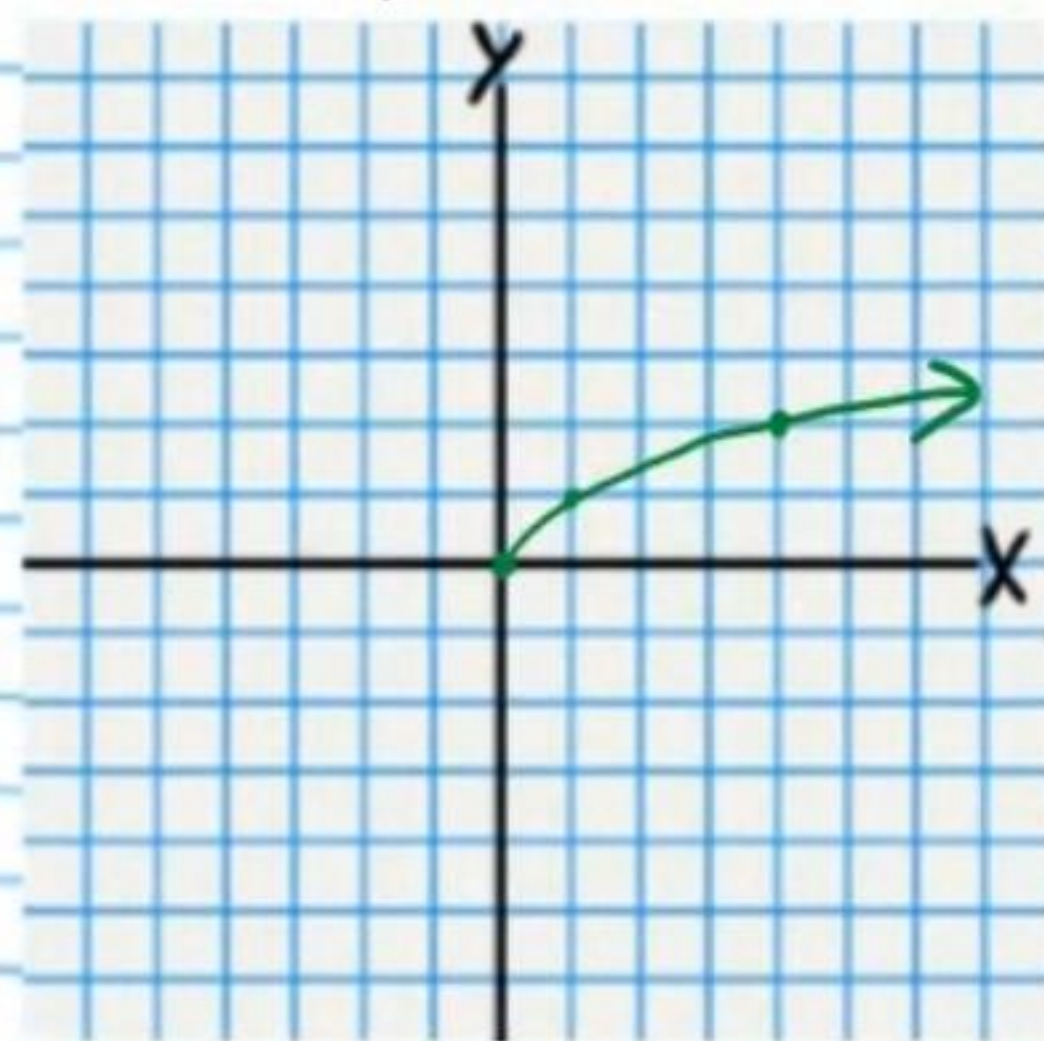
$$f(x) = 3\sqrt{x}$$

$$y = 3\sqrt{x}$$

$$\frac{y}{3} = \sqrt{x}$$

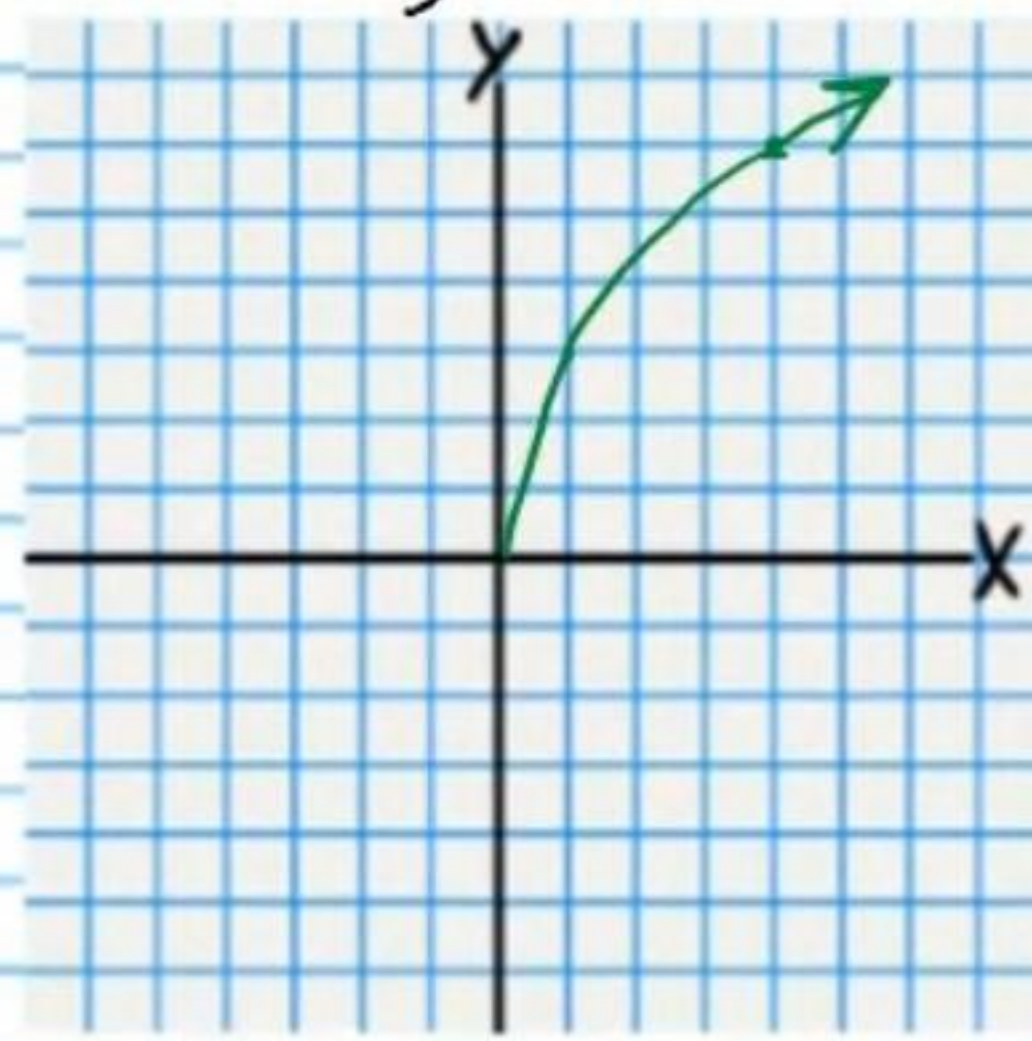
Parent Function

$$y = \sqrt{x}$$



Transformation

$$\frac{y}{3} = \sqrt{x}$$



ii) Write the domain and range of the function in set notation.

Domain: $\{x \mid x \geq 0\}$

Range: $\{y \mid y \geq 0\}$

⑭ Graph the following quadratic expression. Draw the line of symmetry and write its equation. Write the coordinates of the vertex, x-intercepts, and y-intercept, and draw the points on the graph.

$$y = \frac{1}{2}(x+4)^2 + 1$$

$$y = a(x-h)^2 + k$$

$$a = \frac{1}{2}, h = -4, k = 1$$

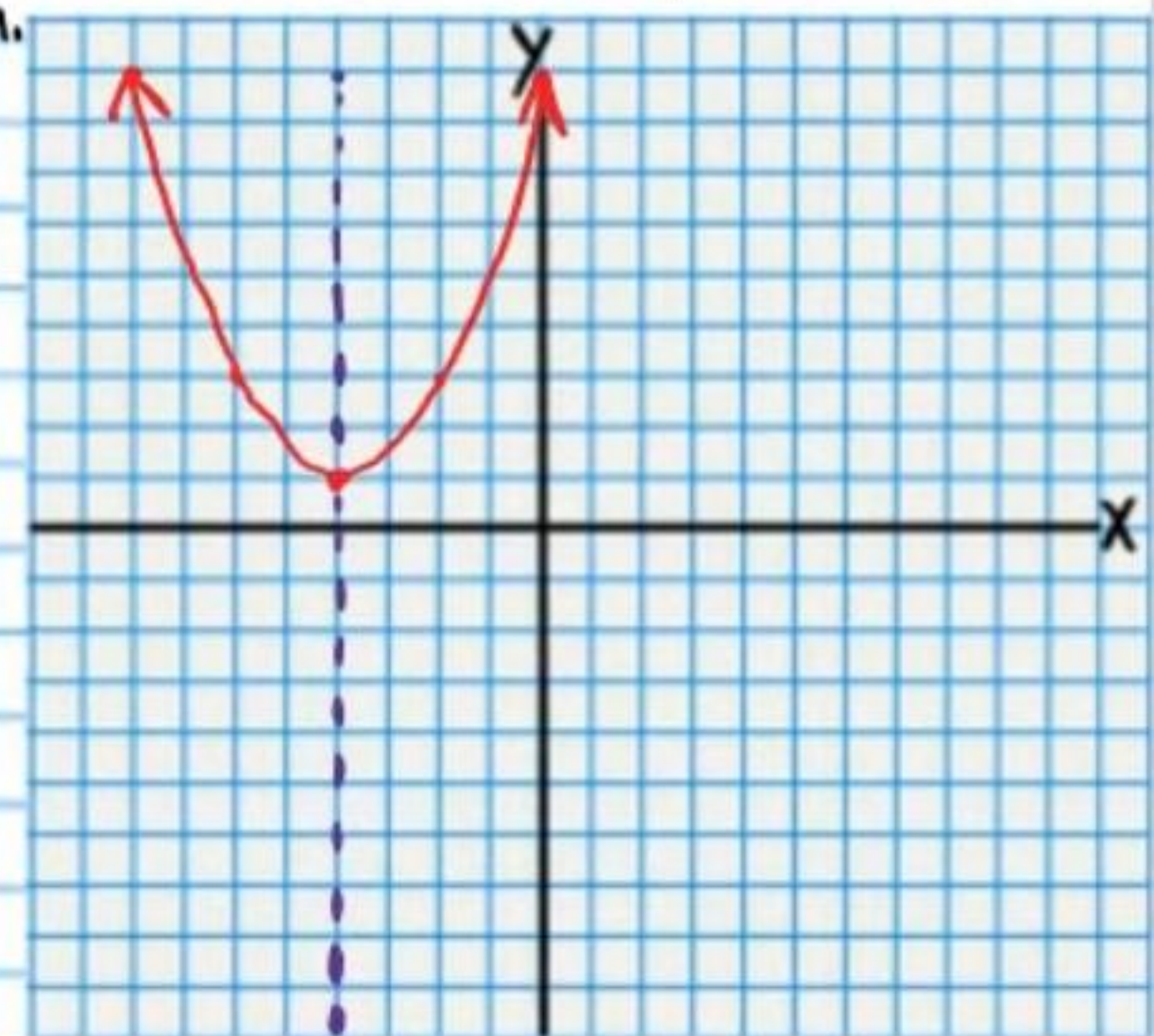
$$V(h, k) = V(-4, 1)$$

No x-intercepts

$$(0, 9)$$

$$V(-4, 1)$$

$$x = -4$$



⑮ Graph the following quadratic expression. Write the coordinates of the vertex, the x-intercepts, and the focus, and draw the points on the graph.

$$-2(x-1)(x+3)=y$$

$$a(x-m)(x-n)=y$$

$$a=-2, m=1, n=-3$$

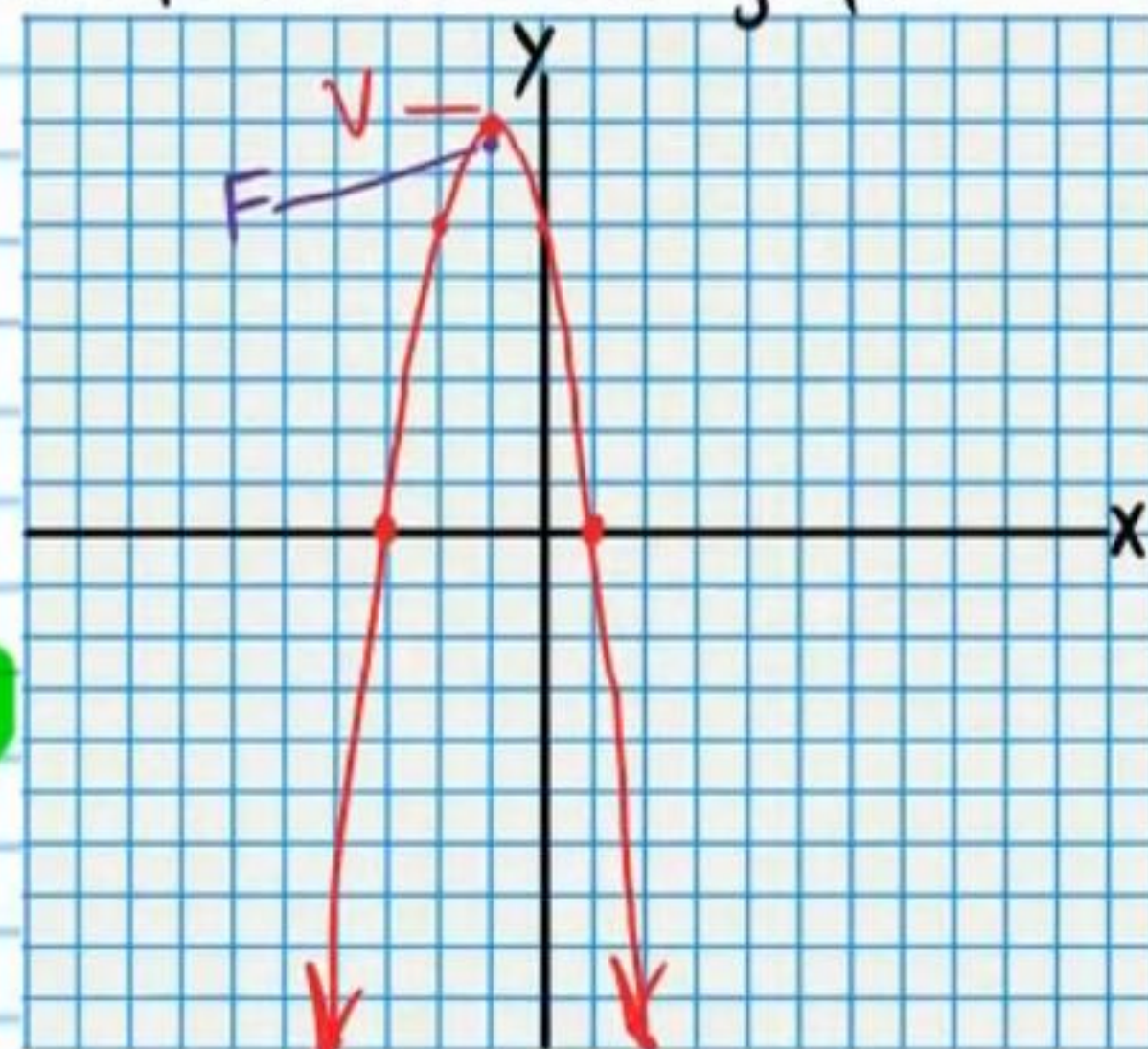
$$V_x = \frac{m+n}{2} = \frac{1+(-3)}{2} = -1$$

$$V_y = -2(-1-1)(-1+3) \\ = -2(-2)(2) \\ = 8$$

$$V(-1, 8)$$

$$p = \frac{1}{4a} = \frac{1}{4(-2)} = -\frac{1}{8}$$

$$F(V_x, V_y+p) = F(-1, 8+(-\frac{1}{8})) = F(-1, 7\frac{7}{8})$$



$$(1, 0) \& (-3, 0)$$

⑮ i) Graph the function below by performing multiple transformations on the graph of the parent function. Graph each transformation and write its equation.

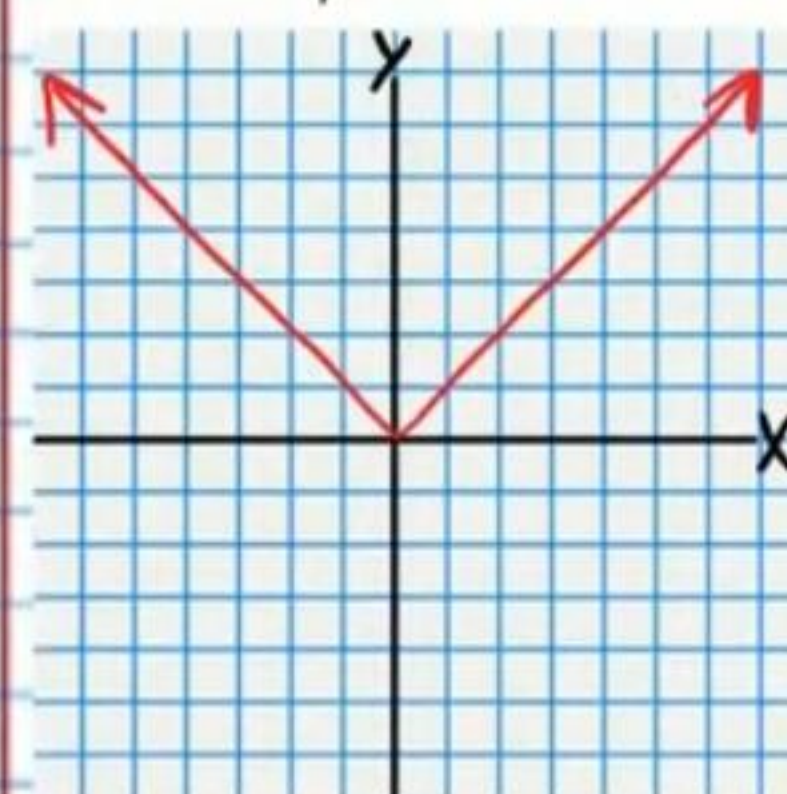
$$f(x) = -|x+3| - 3$$

$$y = -|x+3| - 3$$

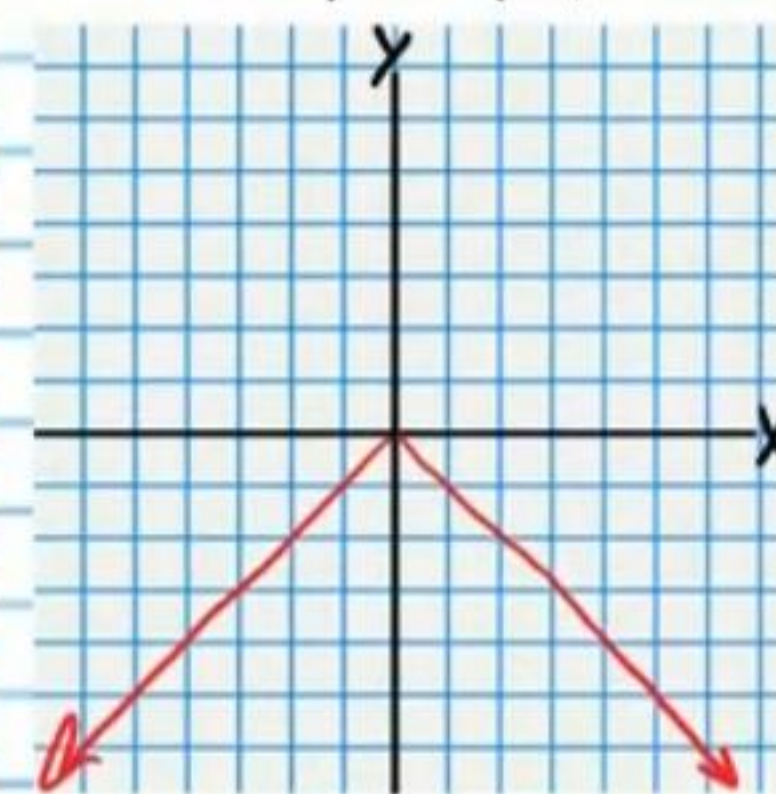
$$y+3 = -|x+3|$$

$$-(y+3) = |x+3|$$

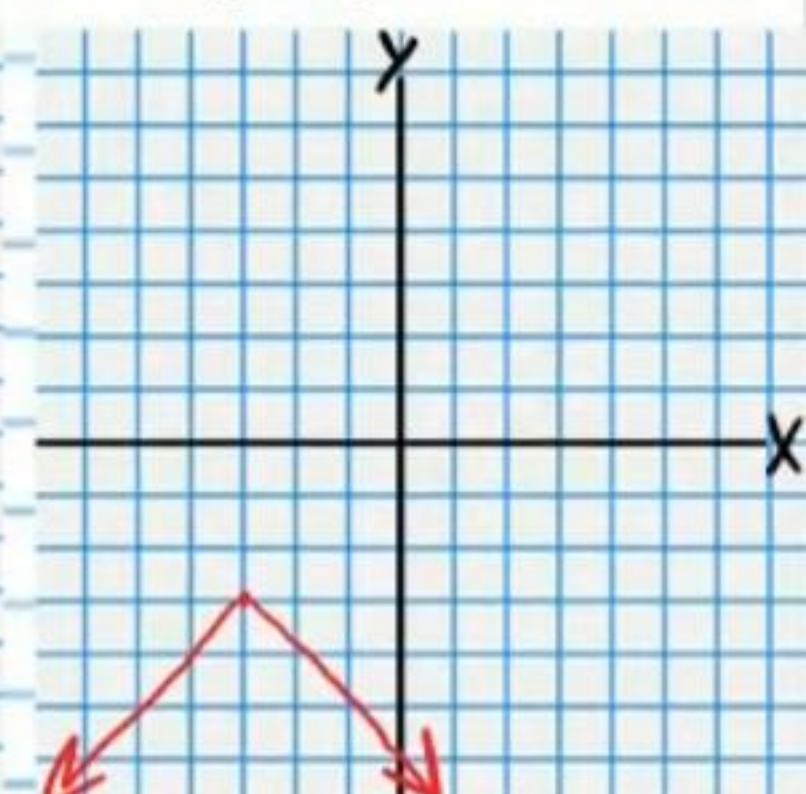
Parent Function
 $y = |x|$



Transformation #1
 $-y = |x|$



Final Transformation
 $-(y+3) = |x+3|$



ii) Write the domain and range of the function in interval notation.

Domain: $(-\infty, \infty)$

Range: $(-\infty, -3]$

⑪ Graph the following quadratic expression. Write the equation of the directrix, the coordinates of the vertex, and the coordinates of the x-intercepts (Write the x-intercepts as exact expressions). Draw the directrix as a dotted line and draw each point on the graph.

$$y = x^2 - 6x + 2$$

$$y = ax^2 + bx + c$$

$$a = 1, b = -6, c = 2$$

$$V_x = -\frac{b}{2a} = -\frac{(-6)}{2(1)} = 3$$

$$V_y = 3^2 - 6(3) + 2 = -7$$

$$V(3, -7)$$

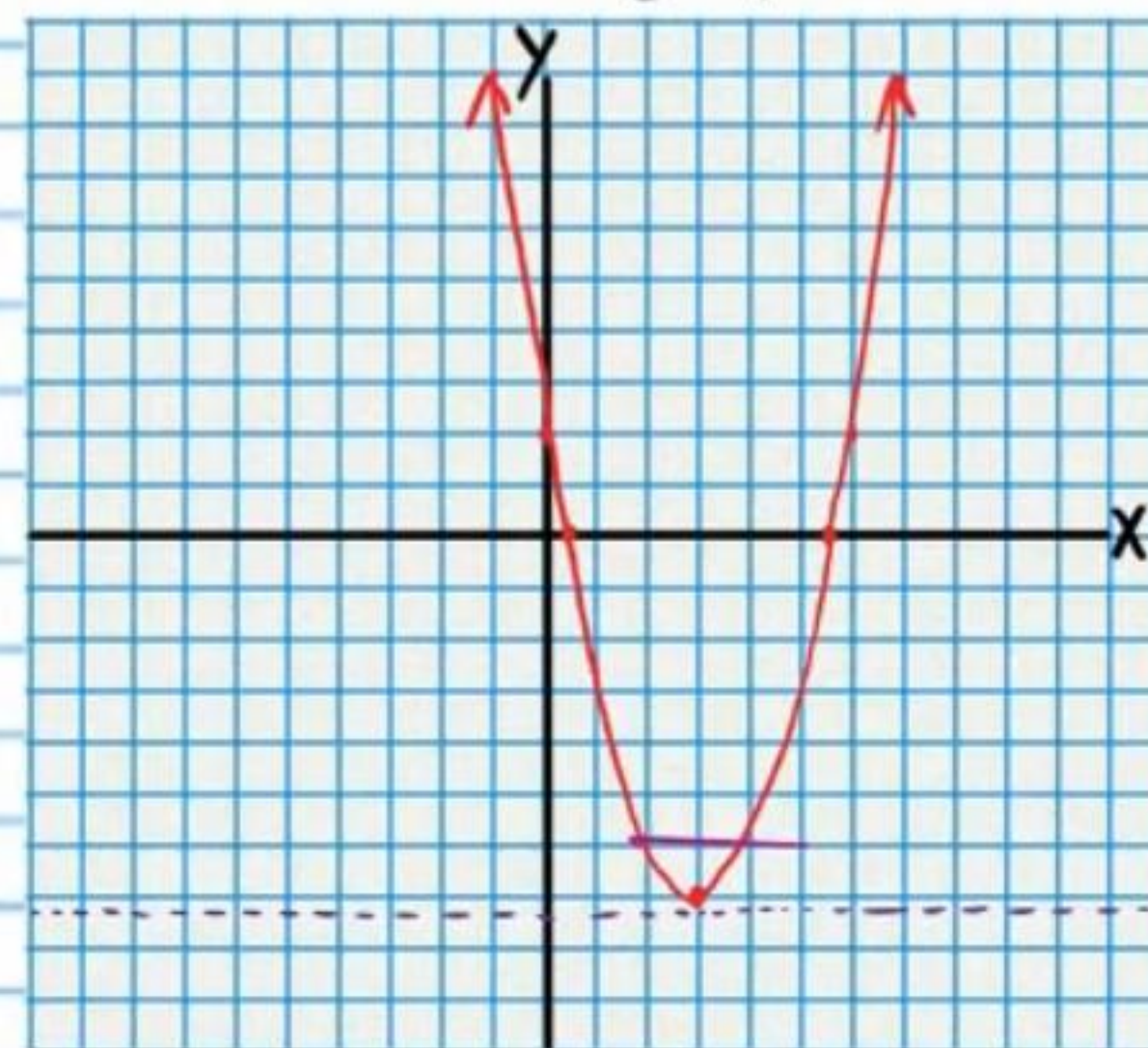
$$0 = x^2 - 6x + 2$$

$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(2)}}{2(1)}$$

$$x = \frac{6 \pm \sqrt{36 - 8}}{2}$$

$$x = \frac{6 \pm \sqrt{28}}{2}$$

$$x = \frac{6 \pm 2\sqrt{7}}{2} = 3 \pm \sqrt{7}$$



$$(3 + \sqrt{7}, 0) \text{ \& } (3 - \sqrt{7}, 0)$$

$$P = \frac{1}{4a} = \frac{1}{4(1)} = \frac{1}{4}$$

$$y = V_y - P$$

$$y = -7 - \frac{1}{4}$$

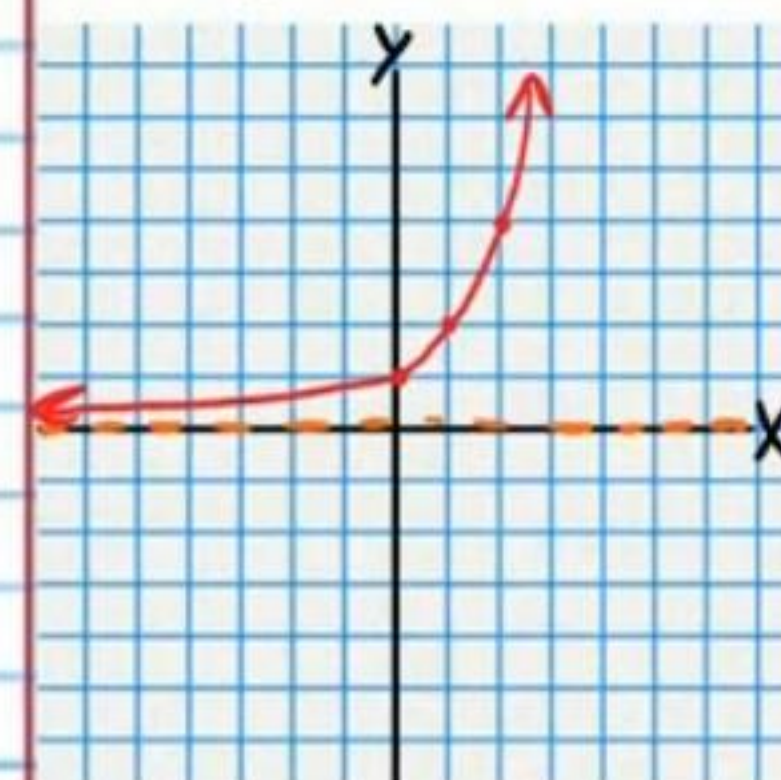
$$y = -7\frac{1}{4}$$

⑫ i) Graph the function below by performing multiple transformations on the graph of the parent function. Graph each transformation and write its equation. Draw the asymptote(s) as dotted lines.

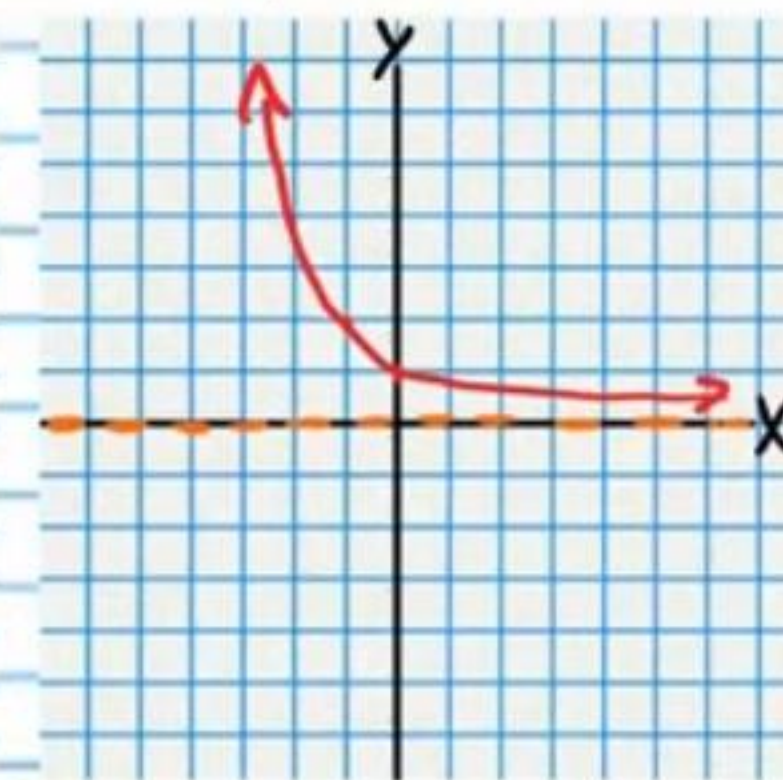
$$f(x) = 2^{-2x}$$

$$y = 2^{-2x}$$

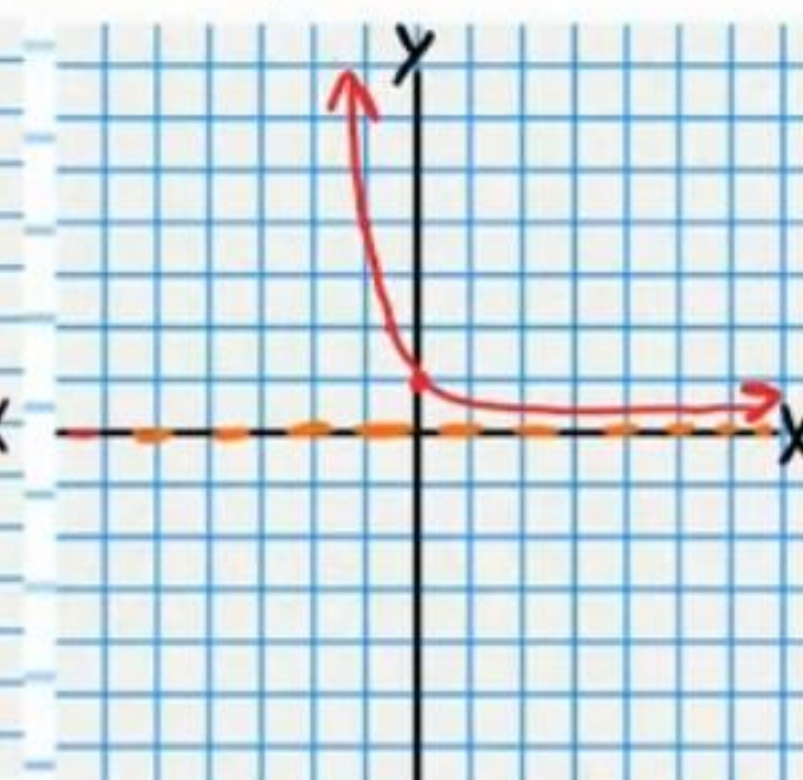
Parent Function
 $y = 2^x$



Transformation #1
 $y = 2^{-x}$



Final Transformation
 $y = 2^{-2x}$



ii) Write the domain and range of the function in set notation.

Domain: $\{\text{All real numbers}\}$

Range: $\{y \mid y > 0\}$

Rationalize and simplify

(19) $\sqrt[3]{\frac{5}{72x^8}}$

$$\frac{\sqrt[3]{5}}{\sqrt[3]{72x^8}}$$

$$\frac{\sqrt[3]{5}}{\sqrt[3]{8 \cdot 9 \cdot x^6 \cdot x^2}}$$

$$\frac{\sqrt[3]{5}}{\sqrt[3]{8} \cdot \sqrt[3]{9} \cdot \sqrt[3]{x^6} \cdot \sqrt[3]{x^2}}$$

$$\frac{\sqrt[3]{5}}{2 \cdot \sqrt[3]{9} \cdot x^2 \cdot \sqrt[3]{x^2}}$$

$$\frac{\sqrt[3]{5}}{2x^2 \sqrt[3]{9x^2}}$$

$$\frac{\sqrt[3]{5}}{2x^2 \sqrt[3]{9x^2}} \cdot \frac{\sqrt[3]{3x}}{\sqrt[3]{3x}}$$

$$\frac{\sqrt[3]{15x}}{2x^2 \sqrt[3]{27x^3}} = \frac{\sqrt[3]{15x}}{2x^2 (3x)} = \boxed{\frac{\sqrt[3]{15x}}{6x^3}}$$

(20) $\frac{\sqrt{2}-1}{\sqrt{6}+2}$

$$\frac{(\sqrt{2}-1)(\sqrt{6}-2)}{(\sqrt{6}+2)(\sqrt{6}-2)}$$

$$\frac{\sqrt{12}-2\sqrt{2}-\sqrt{6}+2}{\sqrt{36}-2\sqrt{6}+2\sqrt{6}-4}$$

$$\frac{2\sqrt{3}-2\sqrt{2}-\sqrt{6}+2}{6+0-4}$$

$$\boxed{\frac{2\sqrt{3}-2\sqrt{2}-\sqrt{6}+2}{2}}$$

(21) Solve by completing the square.

$$3x^2 - 6x - 10 = 0$$

$$x^2 - 2x - \frac{10}{3} = 0$$

$$x^2 - 2x = \frac{10}{3}$$

$$x^2 - 2x + 1 = \frac{10}{3} + 1$$

$$(x-1)(x-1) = \frac{13}{3}$$

$$(x-1)^2 = \frac{13}{3}$$

$$x-1 = \pm \sqrt{\frac{13}{3}}$$

$$x-1 = \pm \frac{\sqrt{39}}{3}$$

$$\boxed{x = 1 \pm \frac{\sqrt{39}}{3}}$$

(22) Convert to the other two forms (i.e. intercept and vertex).

$$y = -x^2 + 3x + 4$$

$$y = -(x^2 - 3x - 4)$$

$$\boxed{y = -(x-4)(x+1)}$$

$$V_x = \frac{m+n}{2} = \frac{4+(-1)}{2} = \frac{3}{2}$$

$$V_y = -\left(\frac{3}{2}\right)^2 + 3\left(\frac{3}{2}\right) + 4$$

$$= -\frac{9}{4} + \frac{9}{2} + 4$$

$$= -\frac{9}{4} + \frac{18}{4} + \frac{16}{4}$$

$$= \frac{25}{4}$$

$$\boxed{y = -\left(x - \frac{3}{2}\right)^2 + \frac{25}{4}}$$

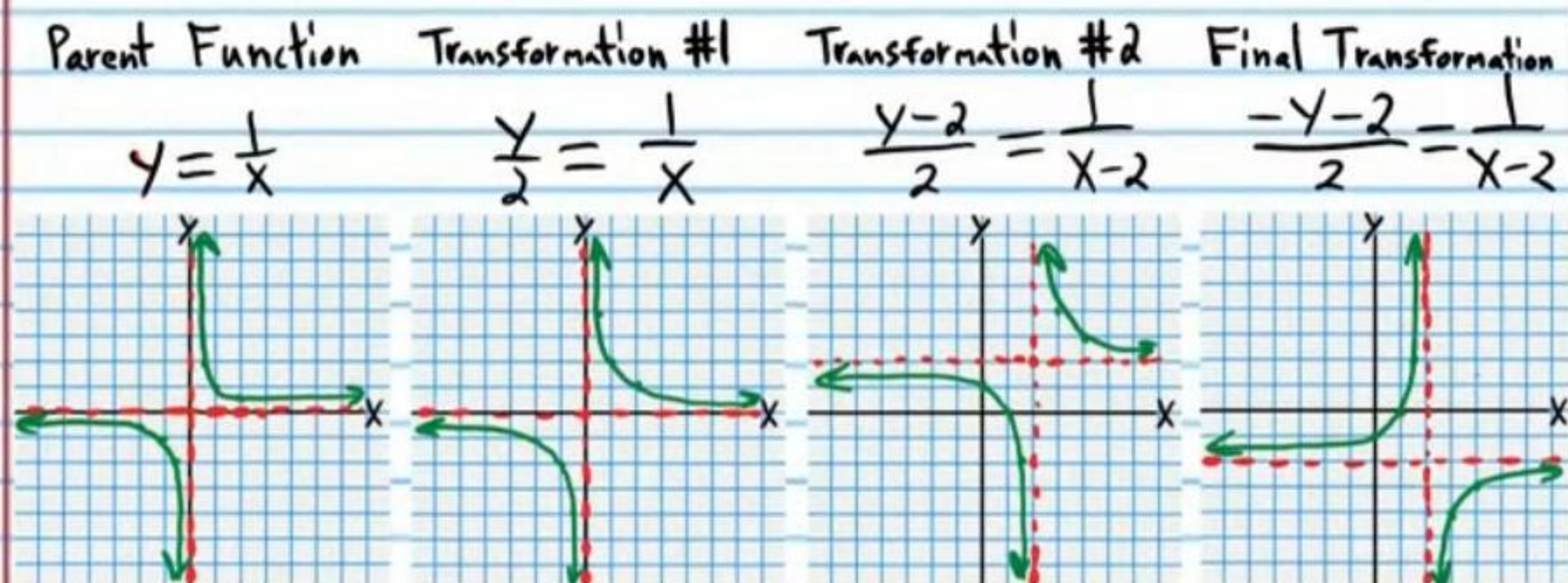
23) i) Graph the function below by performing multiple transformations on the graph of the parent function. Graph each transformation and write its equation. Draw the asymptote(s) as dotted lines.

$$f(x) = -\frac{2}{x-2} - 2$$

$$y = -\frac{2}{x-2} - 2$$

$$y+2 = -\frac{2}{x-2}$$

$$\frac{-y-2}{2} = \frac{1}{x-2}$$



ii) Write the domain and range of the function in interval notation.

Domain: $(-\infty, 2) \cup (2, \infty)$

Range: $(-\infty, -2) \cup (-2, \infty)$

24) i) Money is deposited in an account paying a 2% annual interest rate compounded biannually. After 3 years, the investment grows to \$10,344. How much money was initially deposited in the account? Assume that there were no additional deposits or withdrawals.

$$A = P(1 + \frac{r}{n})^{nt}$$

$$10,344 = P(1 + \frac{0.02}{2})^{2(3)}$$

$$10,344 = P(1.01)^6$$

$$10,344 = P(1.06152)$$

$$P = \$9,744.52$$

ii) If an \$8,000 investment grows to \$12,366 in 20 years, what interest rate was applied if it was compounded continuously?

$$A = Pe^{rt}$$

$$12,366 = 8,000e^{r(20)}$$

$$1.54575 = e^{20r}$$

$$20r = \log_e 1.54575$$

$$20r = \ln(1.54575)$$

$$20r \approx 0.4355$$

$$r \approx 0.0217$$

$$\boxed{2.18\%}$$

(25) A cash register has nickels, dimes, and quarters, but no other types of coins. The total value of the coins is \$5.90. The difference between the number of nickels and the number of dimes is 4. If there are 38 coins in total, how many of each type of coin are there?

Q = # of quarters

N = # of nickels

D = # of dimes

$$Q + N + D = 38$$

$$N - D = 4$$

$$0.25Q + 0.05N + 0.1D = 5.90$$

$$\begin{cases} \times 0.25 \Rightarrow 0.25Q + 0.25N + 0.25D = 9.5 \\ 0.25Q + 0.05N + 0.1D = 5.90 \end{cases}$$

$$\underline{0.2N + 0.15D = 3.6}$$

$$\times 0.2 \begin{cases} 0.2N + 0.15D = 3.6 \\ N - D = 4 \end{cases}$$

$$N - D = 4$$

$$\begin{cases} 0.2N + 0.15D = 3.6 \\ 0.2N - 0.2D = 0.8 \end{cases}$$

$$0.35D = 2.8$$

$$\boxed{D = 8}$$

$$N - 8 = 4$$

$$\boxed{N = 12}$$

$$Q + 12 + 8 = 38$$

$$Q + 20 = 38$$

$$\boxed{Q = 18}$$